

Yixin (Stella) Yuan

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Education

University of California San Diego, La Jolla, CA
Doctor of Philosophy in Computational Neuroscience

GPA 3.99

August 2021 – May 2026

Carnegie Mellon University, Pittsburgh, PA
Bachelor of Science in Neuroscience (Computational Track)

GPA 3.81

August 2015 – May 2019

Experience

Cognitive Neuroscience Researcher, UC San Diego

October 2022 – Present

- Assembled a semi-automated preprocessing pipeline that parses 500GB+ of functional magnetic resonance imaging (fMRI) data; vastly improved data handling efficiency and ensured standardization across projects
- Employed machine learning techniques to reconstruct stimulus features from noisy fMRI signals; created a ridge-regularized circular regression model with R-squared value reaching 0.7
- Developed a feature engineering pipeline based on Gaussian process modeling to enhance signal-to-noise ratio of fMRI data; moderately improved tuning curve estimation compared to generalized additive models
- Adapted the deep neural feature ansatz into RNNs to probe timescale characteristics of feature learning; simulation results complemented fMRI data and was instrumental in guiding experiment planning

Computational Research Fellow, Marcus Autism Center at Emory University

August 2019 – July 2021

- Documented coding standards and procedures for HTSQL queries as a reference manual for team members and leadership; facilitated cross-project communication through integrating pull requests across teams
- Leveraged functional data analysis and factor analysis to parse heterogeneity in Autism Spectrum Disorder; results provided novel insight to how differential gaze patterns predict cognitive development outcomes
- Interfaced with 30+ families per week to coordinate research visits, collect eye-tracking data, and provide guidance through additional clinical assessments

Neurogenomics Student Researcher, Carnegie Mellon University

June 2017 – May 2019

- Compiled a pipeline for quality assessment and genomic-region annotation of preprocessed chromatin immunoprecipitation sequencing (ChIP-seq) data; cut down repetitive manual coding time by roughly 20%
- Leveraged multivariate differential expression and pathway analysis to discover epigenetic biomarkers of aging; revealed cell-type specific regulatory mechanisms linked to pathogenesis of Alzheimer's disease

Neurolinguistics Research Intern, Tokyo Institute of Technology, Japan

June 2018 – August 2018

- Pilot tested a novel linguistics fMRI project via audio stimulus generation, data preprocessing and multivariate pattern analysis; organized scripts and provided feedback which improved experimental design for the main study
- Assessed functional connectivity of the brain and tested against meta-analysis leveraging the Neurosynth database; results disclosed separate brain networks contributing to supervised versus non-supervised speech segmentation

Skills

- **Programming:** 8+ years of project management and coding experience in R, Matlab and Python. Additional proficiency in HTSQL, C++, html, CSS and Java
- **Data Analysis:** 7+ years of making data-driven research decisions leveraging dimensionality reduction, cluster analysis, regression analysis, Gaussian process modeling and Bayesian optimization
- **Deep Learning:** extensive experience using TensorFlow and PyTorch to build deep neural networks including CNNs, RNNs and variational autoencoders
- **Effective Communication:** excellent interpersonal and presentation skills, having interfaced with 100+ scientific audience when presenting at 3 international conferences. Outstanding critical thinking and idea synthesis ability, with research projects culminating in 3 scientific publications